

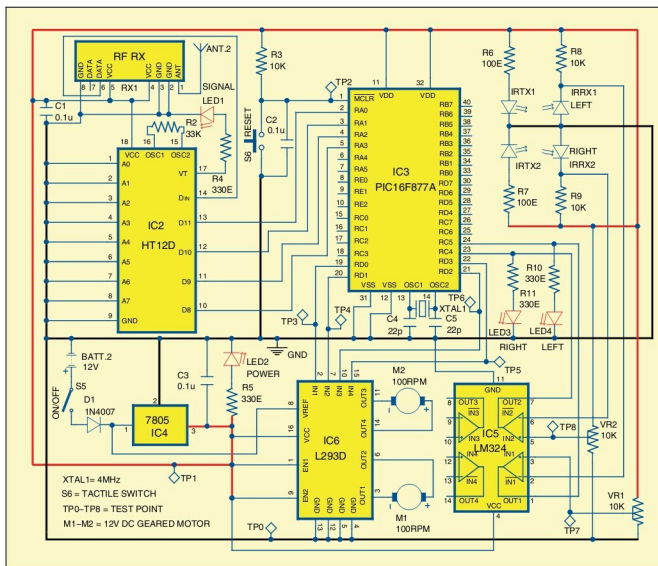


Fig. 3: Circuit of robot

Regulator IC4 provides 5V regulated supply. 12V battery supply is directly connected to pin 8 of motor driver IC6. Two IR transmitter-receiver pairs are used for obstacle detection. Presets VR1 and VR2 are used to set the reference voltage for the two operational amplifiers in IC5.

When the robot reaches close to an obstacle, the reflected IR beam is detected. Voltage at the inverting pins of operational amplifiers goes lower than the reference voltage and their outputs go high. The same is indicated by LED3 and LED4. Output pins 1 and 7 of IC5 are connected to pins RC5 and RC4 of microcontroller IC3. The microcontroller controls the motor con-

responding to the data received from the transmitter and obstacle detection system.

Test Points of Receiver

Test point	Details
TP0	Ground (GND)
TP1	5V
TP2	Reset
TP3	Input for pin 2 of IC6
TP4	Input for pin 7 of IC6
TP5	Input for pin 10 of IC6
TP6	Input for pin 15 of IC6
TP7	Reference voltage for obstacle detection
TP8	Reference voltage for obstacle detection

Working of the system is simple. The robot is in manual mode by default. In manual mode, it can be directed by the remote. For each direction there is a dedicated button on the remote. When the robot encounters any obstacle, it stops moving forward. The robot will, however, be able to move in the rest of the directions.

To make the robot work in automatic mode, press S1 and S2 simultaneously. In automatic mode, the robot will move in forward direction. If it encounters any obstacle, it will stop. To avoid the obstacle, the robot will automatically move a little back, turn to the right by 90 degrees, move forward, then turn left. After the obstacle